

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10 – STATISTICS I (SUB-STRAND 4.1): FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Welcome back to the Maize Harvest Mystery of Nakuru County! A KALRO officer collected the maize harvests (in 90-kg bags) of 20 smallholder farmers. The county government needs ONE number to represent a typical harvest so it can decide how much subsidised fertiliser to allocate next season. Here is the officer's raw list, in the order she collected it: 12, 7, 21, 12, 4, 10, 28, 12, 15, 9, 12, 31, 6, 10, 20, 12, 14, 8, 24, 11. Use everything you learned in this unit to answer the Driving Question: How can we collect, organize, and represent data about our community to help Kenyan decision-makers take effective action? Answer each prompt in the space under it. Use full sentences, show any working, and use the 20 Nakuru harvests above as your evidence. Use the correct statistical words (frequency, tally, mean, median, mode, outlier, axis, sector).

Part 1: From chaos to order

At the start of the unit, the KALRO officer's chalkboard list of 20 harvests looked chaotic. (a) Describe how you would build a frequency distribution table from the raw list, including how tally marks are used. (b) Use the list to build your table, then name one pattern that the table lets you see that the raw list hid. (c) Explain one thing the frequency table still cannot tell the county government.

(a) I would list each different harvest value (from the smallest, 4 bags, to the largest, 31 bags) in one column. Then I would go through the raw list one value at a time and make a tally mark beside the matching value, grouping the marks in fives (four strokes crossed by a fifth) so they are easy to count. Finally I would count the tallies for each value and write the total in a frequency column. (b) My table shows 12 bags with a frequency of 5 (five tallies, one group of five), while 10 bags has a frequency of 2 and every other value occurs once. So 12 bags is the most common harvest, a pattern that is hard to see in the raw list. The table also shows the harvests are spread from 4 bags to 31 bags. (c) The table organises the data but does not give the single number the county needs to represent a typical farmer. We still need a measure of central tendency (mean, median or mode) to do that.

Part 2: Three measures, three stories

Use the 20 Nakuru harvests. (a) Explain in your own words how each of the three measures (mean, mode, median) is found, and then find each one for the 20 harvests, showing your

(a) Mean: add all 20 harvests and divide by 20 ($\bar{x} = \Sigma x \div n$). The total is 278, so the mean is $278 \div 20 = 13.9$ bags. Mode: the value that occurs most often, found by looking for the highest frequency in the frequency table. 12 bags occurs 5 times, so the mode is 12 bags. Median: arrange the 20 values in order (4, 6, 7, 8, 9, 10, 10, 11, 12, 12, 12, 12, 12, 14, 15, 20, 21, 24, 28, 31) and, because 20 is even, average the 10th and 11th values. Both are 12, so the median is 12 bags. (b) The mean of 13.9 bags is the 'fair share' value: if all 278 bags harvested by the 20 farmers were collected and shared out equally, each farmer would get 13.9 bags. It is the balance point of the data. (c) The mean (13.9) is higher than the median (12). A few farmers, such as the ones with

working. (b) Explain what your mean tells us using the idea of a 'balance point' or equal redistribution. (c) Compare your mean with your median. Using the idea of outliers, explain why the two values are different.	28 and 31 bags, have harvests far above the rest of the data. These extreme values are outliers. They pull the mean upward because the mean uses the size of every value, while the median depends only on the middle of the ordered list and stays at 12 bags.
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Part 3: Which number should the county use?

The county government must choose ONE number to decide how much subsidised fertiliser to allocate. Write a short Claim–Evidence–Reasoning advisory note to the Nakuru County Director of Agriculture. Choose mean, median or mode and: (a) state your claim, (b) give evidence from the Nakuru data, (c) explain your reasoning, including how outliers affect your choice, and (d) state one limitation or risk of your choice.	Claim: I recommend that the county use the median of 12 bags as the typical harvest for Nakuru smallholder farmers. Evidence: The median is 12 bags and the mode is also 12 bags (it occurred 5 times out of 20), while the mean is 13.9 bags. The harvests range from 4 to 31 bags. Reasoning: The median is the middle value of the ordered data and is not pulled away by extreme values. The mean is raised by a few farmers with very large harvests (outliers such as 28 and 31 bags), so it is higher than what most farmers actually harvest. The median and mode agree, which shows that 12 bags is what a typical smallholder actually harvests. Using it gives a fairer picture of the farmers who need support. Limitation: Because the median ignores the size of the very low and very high harvests, farmers with only 4 bags may need more help than a single typical number suggests, so the county should also look at the spread of the data (for example, on a graph) before finalising the allocation.
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Part 4: Choosing the right picture

The KALRO officer must present the data to decision-makers. (a) Choose one of the following: a bar graph, a line graph or a pie chart. Describe a question about Kenyan farming or maize that your chosen graph can answer well, and describe how it is constructed (axes, scale, angles or points, as relevant). (b) Name one of the other two graphs and explain why it would be less suitable for the same question. (c) Imagine a bar graph of your frequency table for the 20 harvests, with each harvest	(a) A line graph is best for the question 'How did maize retail prices in Nairobi change from month to month, and when is the best time to sell?' To construct it, I label the horizontal axis with the months (January to December) and the vertical axis with price using an even scale, plot one point for each month, and join consecutive points with straight ruled lines. I give it a title. The graph shows the trend, with peaks and troughs, so the officer can advise farmers when prices are low or high. (Alternative accepted answers: a bar graph to compare harvests across counties such as Nakuru, Uasin Gishu, Trans-Nzoia, Nandi and Kericho, with labelled axes, equal-width bars with gaps and a title; or a pie chart for the share of farmers in each harvest group, with each angle = $(\text{frequency} \div \text{total}) \times 360^\circ$ and angles adding to 360°.) (b) A pie chart would be less suitable for monthly prices because it shows shares of a whole, not change over time, and twelve thin sectors would be hard to read. (c) The tallest bar would be the one for 12 bags, because it has the highest frequency (5). The tallest bar shows the most frequent value, which is the mode.
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value as a category. Which bar would be tallest, and how is it connected to one of the measures of central tendency?	
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Part 5: Answering the Driving Question

Use the whole unit to answer: How can we collect, organize, and represent data about our community to help Kenyan decision-makers take effective action? Write a short paragraph (5–7 sentences) that follows the journey of the Nakuru maize data from the raw list to a decision. Include at least one organising tool, one measure of central tendency and one graph, and say why the choice of each matters.	Statistics is the bridge between raw community information and good decisions. First, the KALRO officer collects the harvests of the 20 Nakuru farmers, but the raw list is chaotic. We organise it in a frequency distribution table with tally marks, which shows that 12 bags is the most common harvest. Next, we calculate the mean (13.9 bags), the mode (12 bags) and the median (12 bags) to find a number that represents a typical farmer. Because a few very large harvests are outliers that pull the mean up, the median is the fairer figure for the county to use when allocating subsidised fertiliser. We then represent the data in a suitable graph, such as a bar graph to compare counties or a line graph to show price changes over time, so that busy decision-makers can understand it quickly. Choosing the wrong measure or graph could mislead the county and leave some farmers without enough support, so careful organisation, calculation and representation lead to effective action for Kenyan farmers.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Organising data with frequency tables	Describes tally marks (grouped in fives), frequency counts and the building of the table accurately; names a pattern revealed and clearly explains what the table cannot answer.	Describes the table and tally marks correctly and names a pattern; the limitation is stated but not fully explained.	Description of the table or tallying is incomplete or confused; patterns or limitations are missing.
Mean, median and mode: methods and meaning	Explains and correctly calculates all three measures, interprets the mean as a balance point, and explains why the mean is above the median using outliers.	Explains the methods correctly with minor gaps in calculation, in interpreting the mean or in the effect of outliers.	Confuses methods or cannot link the values (13.9, 12, 12) to the data.
Evidence-based recommendation (Claim–Evidence–Reasoning)	Makes a clear claim, supports it with exact Nakuru values, reasons about outliers and names a sensible limitation.	Makes a clear claim with some correct evidence and reasoning; limitation is vague.	Claim is unsupported, evidence is missing or inaccurate, or reasoning is absent.
Choosing and constructing graphs	Matches the graph to a suitable question, describes construction correctly (axes, scale, angles or points), justifies why another graph is less suitable and links	Chooses a suitable graph and describes most construction features; the comparison or the mode link is only partly explained.	Graph choice does not fit the question or construction details are missing or wrong.

	the tallest bar to the mode.		
Answering the Driving Question	Coherent paragraph tracing data from collection to decision, using correct statistical vocabulary and connecting tools to Kenyan decision-making.	Covers most steps with generally correct vocabulary; links to decision-making are brief.	Lists tools without connecting them to the data journey or to the decision.

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